MATH200008.01 Optimization
Assignment 3
Due Time: at the beginning of the class, 04/03/2025

1. Provide examples of a function $f : \mathbb{R} \to \mathbb{R}$ and a nonempty subset C of $\mathbb{R}$ illustrating each of the following:

   (i) f is not convex, C is convex, and f is convex on C.

   (ii) f is not convex, C is not convex, and f is convex on C.

2. Let $f : \mathbb{R} \to (-\infty, +\infty]$ be convex with $\operatorname{int} \operatorname{dom} f \neq \varnothing$. Show that f is strictly convex if and only if it is strictly convex on $\operatorname{int} \operatorname{dom} f$.

3. Let $f : \mathbb{R}^n \to (-\infty, +\infty]$. Show that f is convex if and only if

$$(\forall x, y \in \operatorname{dom} f) \quad x \neq y \quad \Rightarrow \quad (\forall z \in (x, y)) \quad \frac{f(z) - f(x)}{\|z - x\|} + \frac{f(z) - f(y)}{\|z - y\|} \leqslant 0.$$

4. Let $\psi : \mathbb{R} \to \mathbb{R}$ be a continuous increasing function such that $\psi(0) \geqslant 0$, and set

$$f : \mathbb{R}^n \to \mathbb{R} : x \mapsto \int_0^{\|x\|} \psi(t)\, dt.$$

   Then the following hold:

   (i) f is convex.

   (ii) Suppose that ψ is strictly increasing. Then f is strictly convex.

5. Consider the Huber robust regression problem for a design matrix $A \in \mathbb{R}^{m \times n}$ with rows $a_i^\top$ and a response vector $b \in \mathbb{R}^m$:

$$\min_{x \in \mathbb{R}^n} \sum_{i=1}^{m} H_\delta(a_i^\top x - b_i)$$

   where the Huber loss function with threshold $\delta > 0$ is defined as:

$$H_\delta(r) = \begin{cases} \frac{1}{2} r^2 & \text{if } |r| \leq \delta \\ \delta |r| - \frac{1}{2}\delta^2 & \text{if } |r| > \delta \end{cases}$$

   By introducing auxiliary variables to decompose the residuals, reformulate this non-smooth problem into a standard constrained Quadratic Program (QP). Clearly state the new decision variables, the objective function, and all constraints.

6. Let $a_1, a_2, \ldots, a_m \in \mathbb{R}^n$. Consider the following unconstrained optimization problem:

$$\min_{x \in \mathbb{R}^n} \quad f(x) = \sum_{i=1}^{m} \|a_i - x\|_2^2$$

(a) Find the analytical optimal solution x^* to this problem.

(b) Determine whether this is a Second-Order Cone Program (SOCP). If it is, explicitly write down the reformulated problem in standard SOCP form.